

# Barrier Lyapunov Function

#Explanation

#Chairing

**Barrier Lyapunov Functions (BLF)** enforce **state constraints** by designing the **Lyapunov Function**  $V_B$  to grow unbounded as  $x$  approaches the constraint boundary, effectively making sure that the system **never violates constraints**. For our system, we define our **constraint** as:

$$\mathbf{x} = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}$$
$$\|\mathbf{x}\|_P^2 < x^+$$

^0f3d2e

The constant  $x^+$  is defined as the boundary of the state, in the case of two **cobots** working on the same **workspace**, this constant is constantly updated by the state of the other robot. This **dynamic constraint** is defined:

$$\|\mathbf{x}\|_P^2 < x^+(\mathbf{x}_j), \quad i, j \in \{1, 2\}, \quad i \neq j$$

Where  $x^+(\mathbf{x}_j)$  is updated based on the other robot's state to avoid collisions.

## Barrier Lyapunov Function

From our [Euler-Lagrange Formulation](#), we can get our *joints' acceleration* as:

$$\ddot{q} = B^{-1}(q)(\tau - C(\dot{q}, q)\dot{q} - G(q)) + B^{-1}\tau + \eta$$

We can propose a **PD controller** that compensates for the dynamics such as:

$$u = BK_p q + BK_D \dot{q} + C(q, \dot{q})\dot{q} + G(q)$$

After substituting, we get:

$$\ddot{q} = K_p q + K_D \dot{q} + \eta$$

Where  $\eta$  represents **unmodeled dynamics/external perturbations**. In [state-space form](#):

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & I \\ K_p & K_D \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ I \end{bmatrix} \eta$$
$$= A\mathbf{x} + B\eta$$

## Structure of BLF

The proposed *Barrier Lyapunov Function* is:

$$V_B = \ln \left( \frac{x^+}{x^+ - ||\mathbf{x}||_P^2} \right) + \lambda_1(t) \text{tr}\{\tilde{K}_P \tilde{K}_P^T\} + \lambda_2(t) \text{tr}\{\tilde{K}_D \tilde{K}_D^T\}$$

Where:

$\tilde{K}_P = K_P - K_P^0$  : Proportional parameter estimation error.

$\tilde{K}_D = K_D - K_D^0$  : Derivative parameter estimation error.

$\lambda_{1,2}(t)$  : Time-varying adaptive gains.

$r$  : Adjusts the decay rate of the decaying gain.

The logarithmic term  $\ln \left( \frac{x^+}{x^+ - ||\mathbf{x}||_P^2} \right)$  becomes singular as the *state* approaches the boundary  $x^+$ , ensuring our [boundary condition](#). The [trace terms](#)  $\text{tr}\{\tilde{K}_P \tilde{K}_P^T\}$  &  $\text{tr}\{\tilde{K}_D \tilde{K}_D^T\}$  are scalars penalizing **estimation errors** in the adaptive controller, and is equivalent to the [Squared Frobenius norm](#)  $||\tilde{K}_P||_F^2$ .

The **adaptive gain term**  $\lambda_{1,2}(t)$  is defined as:

$$\lambda_i = \lambda_{i,0} \left( \frac{x^+ - ||\mathbf{x}||_P^2}{x^+} \right)^r, r > 0, i = 1, 2$$

Where the exponent  $r$  tunes how *sharply* adaptation **decays** near the *boundary*; and  $\lambda_{i,0}$  represents a **constant** baseline **adaptation rate** for when the system **is far** from the *constraints*. Formally, as:

$$||\mathbf{x}||_P^2 \ll x^+$$

Then:

$$x^+ - ||\mathbf{x}||_P^2 \approx 1$$

So our term adapts **fully**:

$$\lambda_i(t) \rightarrow \lambda_{i,0} \cdot 1^r = \lambda_{i,0}.$$

On the contrary, when the system **approaches** the *constraint*:

$$||\mathbf{x}||_P^2 \rightarrow x^+$$

Then:

$$x^+ - ||\mathbf{x}||_P^2 \rightarrow 0$$

So adaptation is **suppressed** to prioritize **safety**:

$$\lambda_i(t) \rightarrow \lambda_{i,0} \cdot 0^r = 0$$

In summary:

| State               | Distance to Boundary            | Adaptive Gain $\lambda_i(t)\lambda_i(t)$ | Barrier Term                |
|---------------------|---------------------------------|------------------------------------------|-----------------------------|
| Far from constraint | $x^+ - \ x\ _P^2 \approx x^+$   | $\lambda_i(t) \approx \lambda_{i,0}$     | Negligible ( $\ln(1) = 0$ ) |
| Near constraint     | $x^+ - \ x\ _P^2 \rightarrow 0$ | $\lambda_i(t) \rightarrow 0$             | Dominant ( $\ln(\infty)$ )  |

## Derivative of the BLF

From its definition, its clear that the proposed **BLF** covers the first two requirements of any **Lyapunov Function**. Now, its derivative is derived as follows:

$$\frac{d}{dt}V_B = \frac{d}{dt}\ln\left(\frac{x^+}{x^+ - \|x\|_P^2}\right) + \frac{d}{dt}(\lambda_1(t)\text{tr}\{\tilde{K}_P\tilde{K}_P^T\}) + \frac{d}{dt}(\lambda_2(t)\text{tr}\{\tilde{K}_D\tilde{K}_D^T\})$$

Where:

$$\begin{aligned}\frac{d}{dt}\ln\left(\frac{x^+}{x^+ - \|x\|_P^2}\right) &= \cancel{\frac{d}{dt}\ln(x^+)} - \frac{d}{dt}\ln(x^+ - \|x\|_P^2) \\ &= -\frac{1}{x^+ - \|x\|_P^2} \frac{d}{dt}(\cancel{x^+} - 2x^T Px) \\ &= \frac{2}{x^+ - \|x\|_P^2} x^T P \dot{x}\end{aligned}$$

And given the derivative of the trace is:

$$\frac{d}{dt}\text{tr}\{\tilde{K}_P\tilde{K}_P^T\} = 2\text{tr}\{\tilde{K}_P^T \dot{\tilde{K}}_P\}$$

Then:

$$\begin{aligned}\frac{d}{dt}(\lambda_1(t)\text{tr}\{\tilde{K}_P\tilde{K}_P^T\}) &= \dot{\lambda}_1(t)\text{tr}\{\tilde{K}_P\tilde{K}_P^T\} + 2\lambda_1(t)\text{tr}\{\tilde{K}_P^T \dot{\tilde{K}}_P\} \\ \frac{d}{dt}(\lambda_2(t)\text{tr}\{\tilde{K}_D\tilde{K}_D^T\}) &= \dot{\lambda}_2(t)\text{tr}\{\tilde{K}_D\tilde{K}_D^T\} + 2\lambda_2(t)\text{tr}\{\tilde{K}_D^T \dot{\tilde{K}}_D\}\end{aligned}$$

So finally:

$$\begin{aligned}\frac{d}{dt}V_B &= \frac{2\mathbf{x}^T P \dot{\mathbf{x}}}{x^+ - \|\mathbf{x}\|_P^2} + \dot{\lambda}_1(t) \text{tr}\{\tilde{K}_P \tilde{K}_P^T\} + 2\lambda_1(t) \text{tr}\{\tilde{K}_P^T \dot{\tilde{K}}_P\} + \dot{\lambda}_2(t) \text{tr}\{\tilde{K}_D \tilde{K}_D^T\} + 2\lambda_2(t) \text{tr}\{\tilde{K}_D^T \dot{\tilde{K}}_D\} \\ &= \frac{2\mathbf{x}^T P \dot{\mathbf{x}}}{x^+ - \|\mathbf{x}\|_P^2} + \text{tr}\{(2\lambda_1 \dot{\tilde{K}}_P + \dot{\lambda}_1(t) \tilde{K}_P) \tilde{K}_P^T\} + \text{tr}\{(2\lambda_2 \dot{\tilde{K}}_D + \dot{\lambda}_2(t) \tilde{K}_D) \tilde{K}_D^T\}\end{aligned}$$

We can [substitute](#)  $\dot{\mathbf{x}}$  for  $A\mathbf{x} + B\eta$ :

$$= \underbrace{\frac{2\mathbf{x}^T P (A\mathbf{x} + B\eta)}{x^+ - \|\mathbf{x}\|_P^2}}_{\text{System Dynamics Term}} + \text{tr}\{(2\lambda_1 \dot{\tilde{K}}_P + \dot{\lambda}_1(t) \tilde{K}_P) \tilde{K}_P^T\} + \text{tr}\{(2\lambda_2 \dot{\tilde{K}}_D + \dot{\lambda}_2(t) \tilde{K}_D) \tilde{K}_D^T\}$$

## Simplifying the Dynamics Term

We can expand  $A = \begin{bmatrix} 0 & I \\ K_P & K_D \end{bmatrix}$  and  $B = \begin{bmatrix} 0 \\ I \end{bmatrix}$ :

$$2\mathbf{x}^T P (A\mathbf{x} + B\eta) = \underbrace{2\mathbf{x}^T P A \mathbf{x}}_{\text{Nominal}} + \underbrace{2\mathbf{x}^T P B \eta}_{\text{Perturbative}}$$

$$2\mathbf{x}^T P \begin{bmatrix} 0 \\ I \end{bmatrix} [\tilde{K}_P \quad \tilde{K}_D \mathbf{x}] = 2\mathbf{x}^T P \begin{bmatrix} 0 \\ I \end{bmatrix} [I \quad 0] \tilde{K}_P \mathbf{x} + 2\mathbf{x}^T P \begin{bmatrix} 0 \\ I \end{bmatrix} [0 \quad I] \tilde{K}_D \mathbf{x}$$

$$\begin{bmatrix} \mathbf{x} \\ \eta \end{bmatrix}^T \begin{bmatrix} PA + A^T P + Q & PB \\ B^T P & -\epsilon I \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \eta \end{bmatrix}$$